

Reza Adinepour

Department of Computer Engineering,
Tehran Polytechnic,
Tehran, Iran

Homepage: <https://rezaadinepour.github.io/>
E-mails: adinepour@aut.ac.ir
r3zaadinepour@gmail.com

RESEARCH INTERESTS

- ◇ AI Hardware Accelerators
- ◇ Real-time and Embedded Systems
- ◇ Reconfigurable Computing
- ◇ Neuromorphic Computing
- ◇ Parallel and Distributed Systems
- ◇ Cyber-Physical Systems (CPS)

EDUCATION

M.Sc. in Computer Engineering, Sep. 2023 - Present
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

- Thesis: “*FPGA-Based Hardware Acceleration of Remaining Useful Life Prediction of Rotating Machinery Using Transformer Neural Network*”
- Advisor: [Prof. Morteza Saheb Zamani](#)
- GPA: 3.4/4

B.Sc. in Electrical Engineering, Sep. 2019 - Jun. 2023
Shahrood University of Technology, Shahrood, Iran

- Thesis: “*Design Real Time Face Recognition Systems Based on LBP Features on ODROID-XU4 Embedded Computer Board*”
- Advisor: [Prof. Alireza Ahmadyfard](#)
- GPA: 3.4/4

PUBLICATIONS

- ◇ **Reza Adinepour** and Morteza Saheb Zamani. “*Real-Time RUL Prediction via a Persistent-Weight Dual-Kernel Transformer Accelerator on FPGA*” IEEE Transactions on Computers Journals. IEEE, 2026. (Under-Review)
- ◇ **Reza Adinepour**, Shayan Naghizadeh and Morteza Saheb Zamani. “*Edge Deployment of Quantized Transformer Models for Remaining Useful Life Prediction*” The 34th International Conference on Electrical Engineering (ICEE 2026). IEEE, 2026.
- ◇ Shayan Naghizadeh, **Reza Adinepour** and Morteza Saheb Zamani. “*Low-Precision POSIT Arithmetic for Spiking Neural Networks with Kahan Summation*” The 11th International Conference on Signal Processing and Intelligent Systems (ICSPIS 2025). IEEE, 2025.

RESEARCH COLLABORATIONS

- ◇ **FPGA-Based Hardware Acceleration of Vision Transformer (ViT)** Aug. 2024 - Jun. 2025
Research Assistant, Supervisor: [Prof. Elif Bilge Kavun](#), Department of Computer Engineering, [Dresden University of Technology](#).
 - *Studies and research focused on **Vision Transformer hardware acceleration***
I was conducting research on the implementation and acceleration of ViT on FPGA with the goal of Deep fake image generation.

TEACHING EXPERIENCE

Teaching Assistant-Amirkabir University of Technology

- **Digital Electronics**  Fall 2025
- **Embedded Systems Modeling & Design**  Spring 2025
- **Digital Logic Design**  Fall 2024

Invited Lecturer-Amirkabir University of Technology

- **Operating System Lab**  Fall 2025
- **Computer Architecture Lab**  Spring 2025
- **Logic Circuits Lab**   Spring 2024 and Fall 2023

Teaching Assistant-Shahrood University of Technology

- **Digital Electronics** Spring 2023
- **Signal and Systems** Spring 2023, Fall 2022, Spring 2022, Fall 2021
- **Analog Electronic** Fall 2022
- **Circuit Theory** Fall 2020, Spring 2020

HONORS AND AWARDS	◇ Direct Admission of Master's Degree at Amirkabir University of Technology (Tehran Polytechnic) ◇ Ranked 2nd (top 1%) in Department of Electrical Engineering, Shahrood University of Technology, Among More Than 120 Students. 2023	
NOTABLE PROJECTS	◇ High-Level to RTL Conversion Framework for CNN Acceleration (In Progress) ◇ Secure and High-Performance Firmware Architecture Customization (In Progress) ◇ FPGA Implementation of Logic Locking in Deep Neural Networks ◇ Research-Oriented SystemC Examples ◇ Algorithm Acceleration on HBM-PIM Architecture using PIMSimulator ◇ FPGA-Based Implementation of CNN Using High Level Synthesis (HLS) ◇ Edge Detector HW/SW Co-design on FPGA ◇ HLS-Based Implementation of Vision Transformer (ViT) ◇ FPGA-Based Implementation of Neural Network	
WORK EXPERIENCE	Member of Digital System Design Automation Laboratory Aug. 2023 - Present Tehran, Iran <i>Job Description:</i> Research Assistant R&D department Member, at D3H-Group Jun. 2023 - Sep. 2023 Al Maryah Island, Abu Dhabi, UAE <i>Job Description:</i> Biomedical Signal Processing Developer R&D department Member, at Radan Electronic StartUp May. 2022 - Aug. 2022 Mashhad, Iran <i>Job Description:</i> Embedded Software Developer	
SKILLS	◇ Programming Languages: ◦ Back-end: C, C++, Rust, Java, Python, Matlab, ◦ HDLs: VHDL, Verilog, HLS, SystemC, Nvidia CUDA, OpenMP ◇ Machine Learning Tools: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas ◇ Applications and Scientific Tools: ◦ FPGA/Embedded Systems Development: Xilinx Vivado, Vitis HLS, Vitis AI, FINN, Xilinx ISE, ModelSim, IAR, Keil, CubeMX, Altium Designer, KiCad, Spice, Arduino IDE ◦ Cloud & DevOps Engineering: Git, GitLab, Docker ◦ Scientific Computing & Research Tools: MATLAB, Gem5 ◇ Operating Systems: Linux, Microsoft Windows ◇ Typesetting: T_EX, L_AT_EX, VIM, Microsoft Word, Gnuplot	
REFERENCES	Prof. Morteza Saheb Zamani Professor, Dept. of Computer Engineering Amirkabir University of Technology Email: szamani@aut.ac.ir Prof. Hamid.R Zarandi Associate Professor, Dept. of Computer Engineering Amirkabir University of Technology Email: h_zarandi@aut.ac.ir	Prof. Mehdi Sedighi Professor, Dept. of Computer Engineering Amirkabir University of Technology Email: msedighi@aut.ac.ir Prof. Hamed Farbeh Assistant Professor, Dept. of Computer Engineering Amirkabir University of Technology Email: farbeh@aut.ac.ir